Fintech Product Growth Analysis

How Early Activation Predicts Retention and Lifetime Value

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Product Analytics Portfolio Project
Tools: PostgreSQL, Python, Pandas, Matplotlib

Executive Summary

This project analyzes how early user activation influences retention and lifetime value. Using transactional data, I found that users who complete two transactions within their first seven days retain more consistently and generate approximately five times higher lifetime value than non-activated users. A simple model suggests that improving activation from 10% to 15% could generate an estimated **$1.19M increase in lifetime revenue**. These results highlight activation as a critical growth lever for product teams.

Problem

Many fintech products require onboarding steps such as identity verification and account setup before users can access the full product experience. While these steps are necessary for security and compliance, they often introduce friction that prevents users from fully engaging with the product.

Understanding which early behaviors lead to long-term engagement is important for product growth. This analysis explores whether early activation behavior predicts stronger retention and higher lifetime value.

Dataset

This analysis uses the **Online Retail II dataset**, which contains transactional records from an e-commerce platform.

Dataset source: *UCI Machine Learning Repository – Online Retail II Dataset.*

To simulate a fintech-style lifecycle:

- Each **customer** is treated as a product user
- Each **invoice** is treated as a product transaction

Key fields used in the analysis include:

- Customer ID
- Invoice Date
- Quantity
- Unit Price
- Country

Tools Used

The analysis was conducted using the following tools:

- PostgreSQL for data cleaning and aggregation
- Python for analysis and visualization
- Pandas for data manipulation
- Matplotlib for visualization

Data Preparation

The dataset required several cleaning steps before analysis.

The following records were removed:

- Cancelled invoices
- Negative quantities (returns)
- Rows with missing customer IDs
- Invalid transaction values

After cleaning the data, the dataset contained:

4,312 unique users

Activation Definition

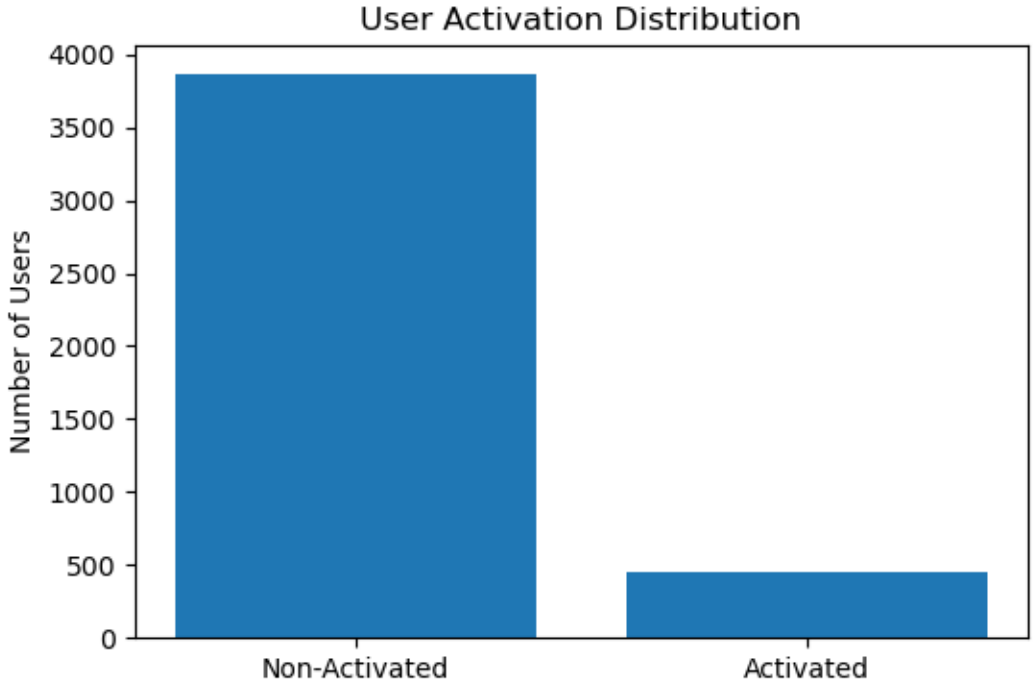
Activation was defined as users completing **at least two transactions within seven days of their first transaction**.

This definition captures meaningful early engagement rather than a single isolated interaction.

Activation Analysis

After applying the activation definition:

- Total users: **4,312**
- Activated users: **442**
- Activation rate: **10.25%**



Only about one in ten users reaches early engagement behavior. This suggests a significant onboarding bottleneck where most users do not progress beyond initial exploration.

Retention Analysis

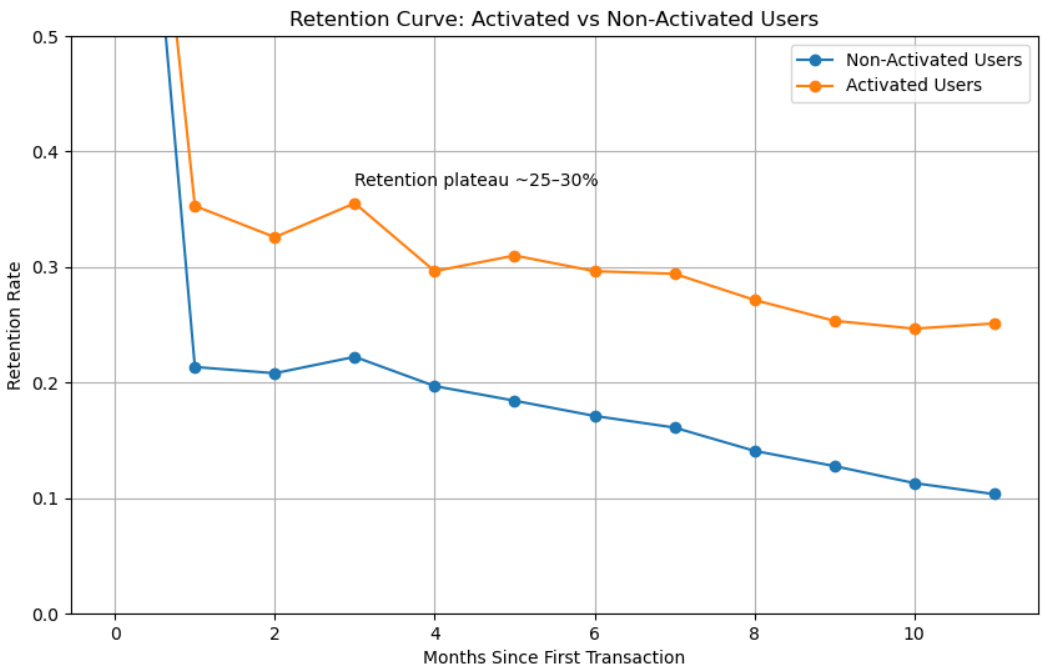
To understand how activation influences long-term engagement, a cohort retention analysis was performed.

30-Day Retention

- Non-activated users: **64%**
- Activated users: **74%**

Activated users were significantly more likely to continue using the product.

Retention Curve



The retention curves reveal a clear behavioral difference between user segments.

Key observations:

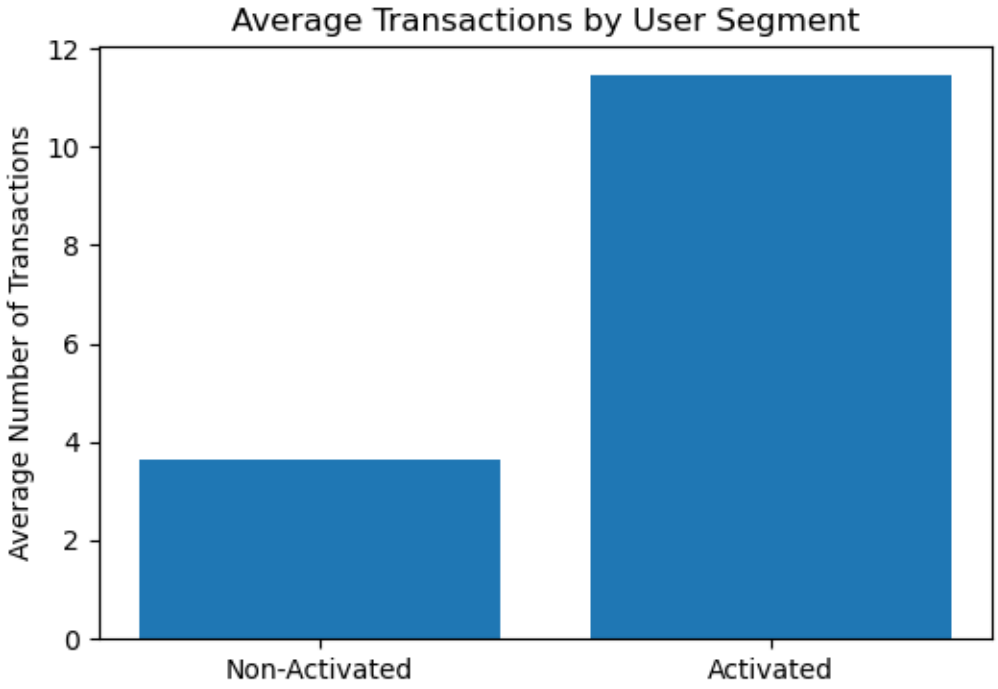
- Activated users retain at higher levels immediately
- Their retention stabilizes around **25–30%**
- Non-activated users steadily decline toward approximately **10% retention**

This indicates that early engagement contributes to long-term habit formation.

User Engagement Comparison

Activated users not only retain longer but also interact with the product significantly more.

Segment	Avg Transactions
Non-activated	3.65
Activated	11.47



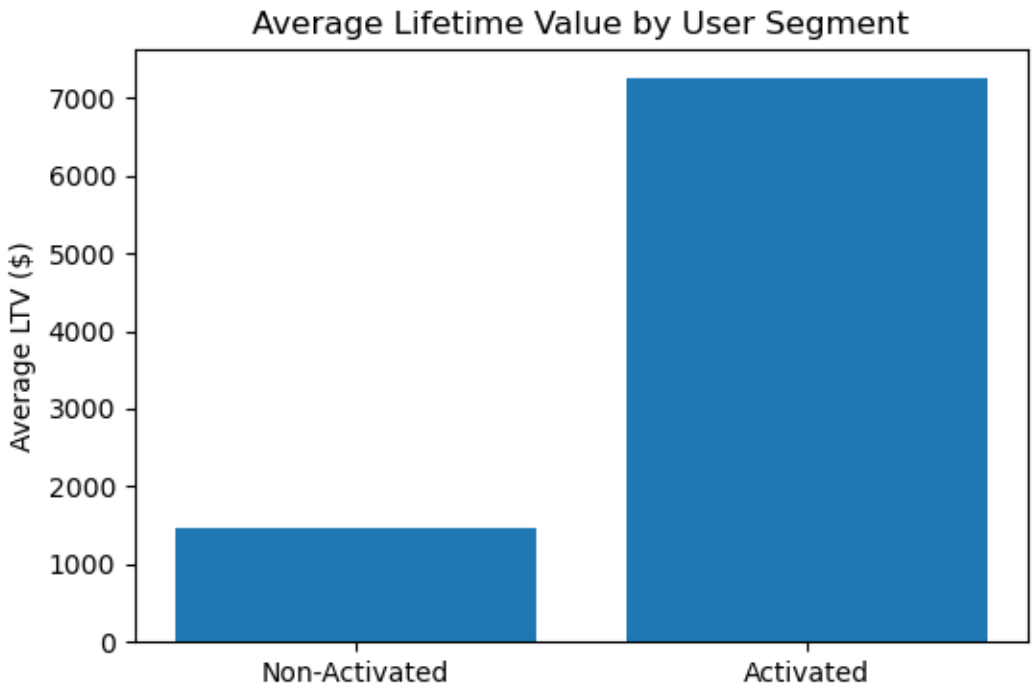
Activated users complete more than **three times as many transactions** as non-activated users, indicating stronger engagement and repeated usage.

The higher engagement levels among activated users help explain why this segment generates significantly higher lifetime value.

Lifetime Value Comparison

Activated users generate substantially higher lifetime value.

Segment	Avg LTV
Non-activated	$1,452
Activated	$7,266



Activated users produce approximately **5× higher lifetime value**, highlighting the economic importance of early engagement.

Business Impact Simulation

To estimate the potential impact of improving activation, a scenario was modeled where activation increases from **10% to 15%**.

Projected results:

- ~205 additional activated users
- Approximately **$1.19M additional lifetime revenue**

This demonstrates that even small improvements in activation could significantly increase long-term revenue.

Experiment Roadmap

Based on the analysis, several experiments could be tested to improve activation.

Increase Trust Before Verification

Introduce trust signals such as security badges, testimonials, or visible user counts to improve confidence during onboarding.

Highlight Product Value Earlier

Show potential benefits before verification, such as financial insights or savings previews.

Simplify Onboarding

Reduce friction by simplifying steps, improving progress indicators, and explaining verification requirements more clearly.

These experiments aim to increase early engagement while monitoring retention and revenue as guardrail metrics.

Limitations

This analysis includes several assumptions:

- Retail transaction data was used to simulate a fintech product lifecycle.
- Activation behavior was approximated using transaction frequency.
- Lifetime value was estimated from transaction totals rather than actual fintech revenue models.

The analysis identifies correlations rather than causal relationships. Controlled experiments would be required to confirm causality.

Key Takeaways

1. Only **10% of users activate early**, indicating a major engagement bottleneck.
2. Activated users generate **5× higher lifetime value**.
3. Activated users form stronger long-term usage habits, stabilizing around **25–30% retention**.

Improving activation represents a **high-leverage opportunity** to increase both retention and revenue.

Final Reflection

Early engagement behavior appears to be a strong predictor of long-term product value. By improving onboarding experiences and encouraging meaningful early interactions, product teams can significantly increase retention, engagement, and lifetime value.